

## Theory:

Bio-electrical signals such as ECG, EMG, and EEG are very low-amplitude signals (ranging from microvolts to millivolts) and are usually accompanied by large amounts of noise, especially common-mode interference. To accurately measure and analyze these signals, a bio-electrical (instrumentation) amplifier is required. Instrumentation amplifier has the following properties: high differential gain, high input impedance, and high common-mode rejection ratio (CMRR). In the first stage, two identical op-amps configured as non-inverting amplifiers, provides very high input impedance, initial amplification of the differential signal and isolation of the signal source from loading effects. Gain of the first stage is given by:

$$A_1 = 1 + \frac{2R_2}{R_1}$$

Any common-mode signal is amplified and preserved for later rejection.

Second stage is a difference amplifier that converts the differential signal into single-ended output. The gain of this stage is

$$A_2 = \frac{R_4}{R_3}$$

This stage is responsible for subtracting the two amplified signals from the first stage and producing the final output voltage. Overall differential gain of the amplifier is the product of the two stage gains, resulting in high gain. This high gain allows very small bio-signals to be amplified to measurable voltage levels.

Common-mode input voltage is defined as the average of two input voltages

$$V_{CM} = \frac{V_1 + V_2}{2}$$

Ideally an instrumentation amplifier should not amplify common-mode signals, and the output should depend only on the differential input ( $V_2 - V_1$ ). However, due to resistor mismatches and non-ideal op-amp behavior, a small amount of common-mode signal appears at the output. The practical output voltage can be written as

$$V_{out} = A_d(V_2 - V_1) + A_{CM}V_{CM}$$

$A_d$  is differential gain and  $A_{CM}$  is common mode gain. Common-mode rejection ratio (CMRR) is defined as

$$CMRR = \frac{A_d}{A_{CM}}$$

### Design Considerations for Biomedical Signals

- For a good EMG amplifier, CMRR should be greater than 80 dB.
- The amplifier gain should remain flat over the signal bandwidth (approximately 10 Hz to 10 kHz for EMG).
- The cut-off frequencies are chosen beyond the signal band to reduce phase distortion.
- The two op-amps in the first stage must be closely matched, preferably from the same IC package (e.g., TL074, LM324).
- High input impedance is required to avoid signal loss due to electrode impedance.

## Data Analysis

For CMRR

Common voltage = 4 V

Finding Vout at different frequencies (Common-mode gain, Acm)

| Frequency | Vout    | Gain Acm = Vout / Vcm |
|-----------|---------|-----------------------|
| 10 Hz     | 5 mV    | $1.25 \times 10^{-3}$ |
| 100 Hz    | 31.2 mV | $7.8 \times 10^{-4}$  |
| 1 kHz     | 15 mV   | $3.75 \times 10^{-3}$ |
| 10 kHz    | 18 mV   | $4.5 \times 10^{-3}$  |
| 100 kHz   | 20 mV   | $5 \times 10^{-3}$    |
| 1 MHz     | 6 mV    | $1.5 \times 10^{-3}$  |
| 5 MHz     | 5 mV    | $1.25 \times 10^{-3}$ |

For  $A_d$  (Differential Gain)

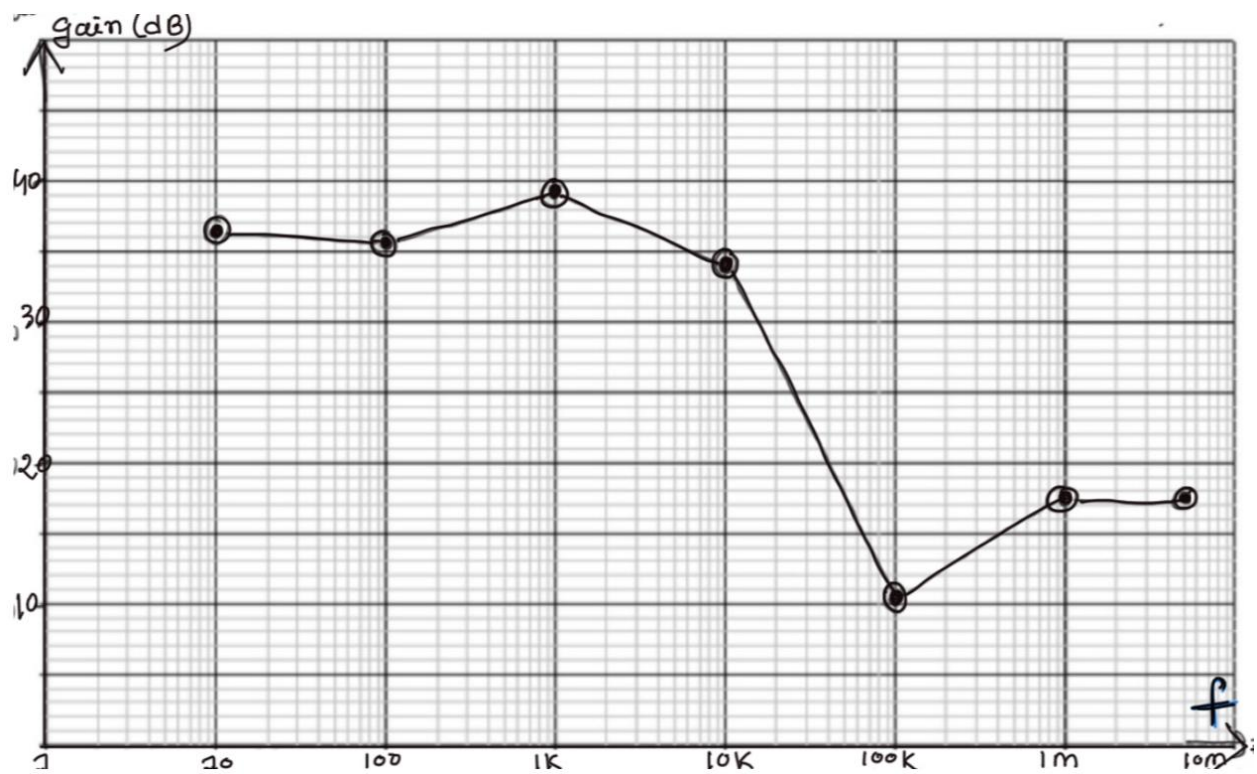
$$V_d = 2 \text{ mV}$$

| Frequency | Vout   | Gain $A_d = V_{out} / V_d$ | Gain (dB) |
|-----------|--------|----------------------------|-----------|
| 10 Hz     | 130 mV | 65                         | 36.25     |
| 100 Hz    | 125 mV | 62.5                       | 35.9      |
| 1 kHz     | 180 mV | 90                         | 39.08     |
| 10 kHz    | 105 mV | 52.5                       | 34.4      |
| 100 kHz   | 6.5 mV | 3.25                       | 10.23     |
| 1 MHz     | 15 mV  | 7.5                        | 17.5      |
| 5 MHz     | 15 mV  | 7.5                        | 17.5      |

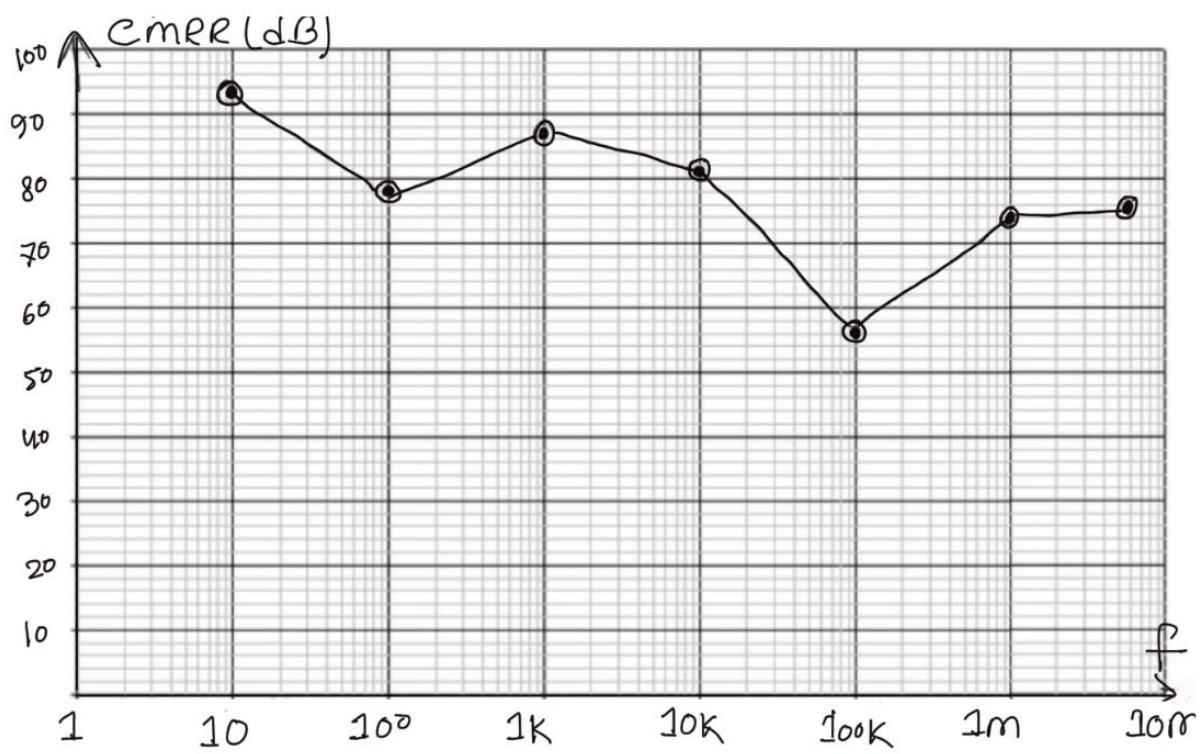
CMRR Calculation

$$CMRR = 20 \log_{10} (A_d / A_{cm})$$

| Frequency | CMRR (dB) |
|-----------|-----------|
| 10 Hz     | 94.72     |
| 100 Hz    | 78.07     |
| 1 kHz     | 87.60     |
| 10 kHz    | 81.33     |
| 100 kHz   | 56.25     |
| 1 MHz     | 73.97     |
| 5 MHz     | 75.56     |



Ad (in dB) vs Frequency (log scale)



CMRR (dB) vs frequency (log scale)

## Discussion:

The instrumentation amplifier studied in this experiment is highly suitable for biomedical signal acquisition, particularly for EMG and ECG measurements. Since bio-electrical signals are extremely weak and susceptible to noise, the high differential gain and high CMRR of the amplifier play a crucial role in ensuring accurate signal amplification. For EMG applications, the amplifier effectively amplifies millivolt-level signals while rejecting large common-mode interference, such as mains hum. Minor improvements, such as adding a notch filter at 50 Hz and a high-pass filter to remove motion artifacts, can further enhance performance. For ECG applications, modifications would be required, including lower gain, narrower bandwidth and improved low-frequency stability to handle baseline wander.

Overall, the experiment successfully demonstrated the working principle, gain characteristics, and importance of CMRR in a bio-electrical instrumentation amplifier. The results confirm that such amplifiers are essential building blocks in modern medical diagnostic systems.